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Trigger menu in 2018

The ATLAS Collaboration

This document summarises the trigger menu deployed by the ATLAS experiment during 2018 data-taking of both proton-proton and lead-lead collisions. Modifications and improvements relative to the proton-proton 2017 trigger menu are discussed, for nominal running conditions of $2.0 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$ and an average of 58 interactions per bunch crossing. Additionally, trigger menus for special proton-proton running configurations are described. The trigger menu for lead-lead collisions is discussed in detail.

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1 Introduction

The trigger system is an essential component of any hadron collider experiment as it is responsible for deciding whether or not to record a given beam crossing for later study. During the second experimental running period (Run 2) of the Large Hadron Collider (LHC), the ATLAS trigger system [1] operated efficiently at proton-proton (pp) collision centre-of-mass energies of $\sqrt{s} = 13$ TeV and $\sqrt{s} = 5.02$ TeV. The LHC operated with instantaneous luminosities of up to $0.5 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$ in 2015, up to $1.4 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$ in 2016, and up to $1.7 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$ in 2017, during 13 TeV pp collisions. A detailed description of the initial performance of the trigger system during 2015 is summarised in Ref. [2]. Summaries of the 2016 and 2017 trigger menus and performance are provided in Ref. [3] and Ref. [4], respectively.

This document summarises the trigger menus used during 2018. Both pp data-taking up to a peak luminosity of over $2.0 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$ and lead-lead (Pb+Pb) data-taking up to a peak luminosity

of $6.2 \times 10^{27} \text{ cm}^{-2}\text{s}^{-1}$ are addressed. Improvements of the trigger system and algorithms with respect to the previously described performance in 2017 [4] are described for pp data-taking.

2 The ATLAS detector and trigger system

The ATLAS detector is a multipurpose particle physics detector with nearly 4π coverage in solid angle around the collision point,¹ and is described in detail in Ref. [5].

Since the start of Run 2, the ATLAS detector has had a two-level trigger system. The first-level trigger (Level-1 trigger, L1) is implemented in hardware and uses a subset of the detector information to reduce the rate of accepted events from an input rate of up to 40 MHz to maximally 100 kHz. The L1 topological trigger processor (L1Topo) [6] combines information about several objects into topological information about the event, and is used to greatly enhance the background rejection at L1 by exploiting the topological features of the targeted signal process.

This is followed by a software-based trigger (high-level trigger, HLT) that reduces the rate of recorded events. An average rate of ~ 1.2 kHz is written out for physics analysis. HLT objects, such as leptons or jets, are only reconstructed when specified L1 triggers have accepted an event. This event reconstruction is referred to as online reconstruction. Events that are accepted by the HLT are written out into different streams and transferred to CERN's Tier-0 computing centre for offline reconstruction.

Event selections in the HLT are referred to as triggers, and the collection of all L1 and HLT triggers and their prescales is called the trigger menu. Triggers are comprised of steps, where a step is a sequence of algorithms. A typical step will execute multiple feature-extraction algorithms requesting event-data fragments from within a rectangular (in $\eta \times \phi$) Region of Interest (RoI) defined by the L1 output, and run reconstruction algorithms over this sub-set of the event data. Each step terminates on a hypothesis algorithm, which uses the reconstructed features to form a boolean decision whether the trigger condition is satisfied. Triggers can also comprise a single hypothesis algorithm that makes no additional selection on an event after the L1 trigger; triggers with this selection are referred to as pass-through. The output of single algorithms and entire steps is cached, as is the partial-event data requested by feature extraction algorithms, and any subsequent requests for the same algorithm/step (within the same RoI) or data fragment(s) within the event processing returns the local cached version. Each L1 and HLT trigger has an associated prescale p_{L1} and p_{HLT} , respectively, which take values ≥ 1 . Only $1/p_{\text{L1}}$ events passing by a L1 trigger accepted for HLT processing and only $1/p_{\text{HLT}}$ events passing the L1 selection required by

¹ ATLAS uses a right-handed coordinate system with its origin at the nominal interaction point (IP) in the centre of the detector and the z -axis along the beam pipe. The x -axis points from the IP to the centre of the LHC ring, and the y -axis points upwards. Cylindrical coordinates (r, ϕ) are used in the transverse plane, ϕ being the azimuthal angle around the z -axis. The pseudorapidity is defined in terms of the polar angle θ as $\eta = -\ln \tan(\theta/2)$. Transverse momentum and energy are defined as $p_T = p \sin \theta$ and $E_T = E \sin \theta$, respectively. Angular distance is measured in units of $\Delta R \equiv \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$.

an HLT trigger are processed at the HLT. Triggers with both $p_{L1} = 1$ and $p_{HLT} = 1$ are referred to as unscaled.

3 Trigger menu for 2018 pp collisions

During the 2018 pp run, the LHC delivered a peak luminosity close to $L = 2.0 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$, and briefly reached $L = 2.1 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$. The runs with peak luminosity were achieved with 2544 colliding bunches, and had a peak number of average interactions per bunch crossing (pileup) of 56 at $L = 2.0 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$ (59 at $L = 2.1 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$), where the average is performed over all colliding bunch pairs. Given the expected luminosity was not significantly higher with respect to the 2017 running conditions, only minimal changes to the trigger menu were required and implemented. This consistency greatly eases the use of the full Run-2 dataset for physics analyses.

Even though the instantaneous luminosity was consistently higher in 2018 than in 2017, when the luminosity was levelled at $1.56 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$, the peak pileup was consistently lower. This was due to the LHC having resolved the issues that enforced a maximum of 1866 colliding bunches, which was compensated for by increasing the number of protons per bunch, and thus the pileup [4]. The rates and/or CPU consumption of some triggers are exceptionally sensitive to the pileup conditions. Thus, at equivalent luminosities, the lower pileup in 2018 reduced the CPU usage and overall output rate, freeing resources for additional developments.

The trigger menu strategy is based on the following building blocks:

- Primary triggers, which are used for physics measurements, and typically run unscaled.
- Support triggers, which are used for efficiency and performance measurements, background estimates or monitoring, and typically run at a small rate (of the order of 0.5 Hz each, with prescales). About 15% of the HLT bandwidth is dedicated to support triggers.
- Alternative triggers, which run alternative online reconstruction algorithms complementary to primary and/or support triggers, and are used, for example, as part of the commissioning of future primary triggers.
- Backup triggers, with tighter selections and lower processing or output rate, in case the CPU usage or output rate of the primary trigger becomes too high.
- Calibration triggers, which are used for detector calibrations, and often run at high rates that are compensated by storing only the necessary detector information for each event, thus reducing the event size and overall bandwidth.

The primary triggers that comprise the 2018 standard physics trigger menu, as optimised for the $1.7 - 2.0 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$ peak instantaneous luminosity regime, and their measured rates are shown in Table 1. The different particle identification (PID) and isolation definitions per object can be found in Ref. [2].

Table 1: The main ATLAS triggers used for runs with luminosities up to $2.0 \times 10^{34} \text{ cm}^{-2} \text{ s}^{-1}$. The observed rates per trigger are reported at a luminosity of $2.0 \times 10^{34} \text{ cm}^{-2} \text{ s}^{-1}$ and a pileup of 55 as raw rates, including rate overlapping between triggers. The uncertainty of the stated rates is typically of the order of a few percent. The total rate corresponds to the full menu, which includes many more triggers than listed in this table. Electron (e), photon (γ), and tau (τ) identification is assumed to be of *medium* efficiency working point, unless specified otherwise. An online isolation requirement is indicated as (i). Topological requirements between L1 objects are indicated as (topo). For the ditau trigger, disambiguation is imposed by a minimal ΔR requirement between isolated L1 tau objects and L1 jet objects. A selection of flavour tagging efficiency ϵ working points are shown for b -jet triggers. For B -physics triggers, dimuon vertices are required to have a positive displacement at the HLT, and dedicated selections for dimuon spectra, J/ψ , B -mesons and Υ are applied as indicated with (di- μ), (J/ψ), (B) and (Υ), respectively. The typical offline selections are only indicative and determined by resolution effects. The total rates at L1 and HLT in the bottom row are the total recorded rates and account for the overlaps between individual triggers. Only a selected number of primary triggers is displayed in this table to indicate the breadth of the physics programme covered by the online trigger selections.

Trigger	Typical offline selection	Trigger Selection		L1 Peak Rate [kHz]	HLT Peak Rate [Hz]
		L1 [GeV]	HLT [GeV]	L= $2.0 \times 10^{34} \text{ cm}^{-2} \text{ s}^{-1}$	
Single leptons	Single isolated μ , $p_T > 27 \text{ GeV}$	20	26 (i)	16	218
	Single isolated tight e , $p_T > 27 \text{ GeV}$	22 (i)	26 (i)	31	195
	Single μ , $p_T > 52 \text{ GeV}$	20	50	16	70
	Single e , $p_T > 61 \text{ GeV}$	22 (i)	60	28	20
	Single τ , $p_T > 170 \text{ GeV}$	100	160	1.4	42
Two leptons	Two μ , each $p_T > 15 \text{ GeV}$	2×10	2×14	2.2	30
	Two μ , $p_T > 23, 9 \text{ GeV}$	20	22, 8	16	47
	Two very loose e , each $p_T > 18 \text{ GeV}$	2×15 (i)	2×17	2.0	13
	One e & one μ , $p_T > 8, 25 \text{ GeV}$	20 (μ)	7, 24	16	6
	One loose e & one μ , $p_T > 18, 15 \text{ GeV}$	15, 10	17, 14	2.6	5
	One e & one μ , $p_T > 27, 9 \text{ GeV}$	22 (e, i)	26, 8	21	4
	Two τ , $p_T > 40, 30 \text{ GeV}$	20 (i), 12 (i) (+jets, topo)	35, 25	5.7	93
	One τ & one isolated μ , $p_T > 30, 15 \text{ GeV}$	12 (i), 10 (+jets)	25, 14 (i)	2.4	17
Three leptons	Three very loose e , $p_T > 25, 13, 13 \text{ GeV}$	$20, 2 \times 10$	$24, 2 \times 12$	1.6	0.1
	Three μ , each $p_T > 7 \text{ GeV}$	3×6	3×6	0.2	7
	Three μ , $p_T > 21, 2 \times 5 \text{ GeV}$	20	$20, 2 \times 4$	16	9
	Two μ & one loose e , $p_T > 2 \times 11, 13 \text{ GeV}$	2×10 (μ)	$2 \times 10, 12$	2.2	0.5
	Two loose e & one μ , $p_T > 2 \times 13, 11 \text{ GeV}$	$2 \times 8, 10$	$2 \times 12, 10$	2.3	0.1
Single photon	One loose γ , $p_T > 145 \text{ GeV}$	24 (i)	140	24	47
Two photons	Two loose γ , each $p_T > 55 \text{ GeV}$	2×20	2×50	3.0	7
	Two γ , $p_T > 40, 30 \text{ GeV}$	2×20	35, 25	3.0	21
	Two isolated tight γ , each $p_T > 25 \text{ GeV}$	2×15 (i)	2×20 (i)	2.0	15
Single jet	Jet ($R = 0.4$), $p_T > 435 \text{ GeV}$	100	420	3.7	35
	Jet ($R = 1.0$), $p_T > 480 \text{ GeV}$	111 (topo: $R = 1.0$)	460	2.6	42
	Jet ($R = 1.0$), $p_T > 450 \text{ GeV}$, $m_{\text{jet}} > 45 \text{ GeV}$	111 (topo: $R = 1.0$)	$420, m_{\text{jet}} > 35$	2.6	36
b -jets	One b ($\epsilon = 60\%$), $p_T > 285 \text{ GeV}$	100	275	3.6	15
	Two b ($\epsilon = 60\%$), $p_T > 185, 70 \text{ GeV}$	100	175, 60	3.6	11
	One b ($\epsilon = 40\%$) & three jets, each $p_T > 85 \text{ GeV}$	4×15	4×75	1.5	14
	Two b ($\epsilon = 70\%$) & one jet, $p_T > 65, 65, 160 \text{ GeV}$	$2 \times 30, 85$	$2 \times 55, 150$	1.3	17
	Two b ($\epsilon = 60\%$) & two jets, each $p_T > 65 \text{ GeV}$	$4 \times 15, \eta < 2.5$	4×55	3.2	15
Multijets	Four jets, each $p_T > 125 \text{ GeV}$	3×50	4×115	0.5	16
	Five jets, each $p_T > 95 \text{ GeV}$	4×15	5×85	4.8	10
	Six jets, each $p_T > 80 \text{ GeV}$	4×15	6×70	4.8	4
	Six jets, each $p_T > 60 \text{ GeV}$, $ \eta < 2.0$	4×15	$6 \times 55, \eta < 2.4$	4.8	15
E_T^{miss}	$E_T^{\text{miss}} > 200 \text{ GeV}$	50	110	5.1	94
B -physics	Two μ , $p_T > 11, 6 \text{ GeV}$, $0.1 < m(\mu, \mu) < 14 \text{ GeV}$	11, 6	11, 6 (di- μ)	2.9	55
	Two μ , $p_T > 6, 6 \text{ GeV}$, $2.5 < m(\mu, \mu) < 4.0 \text{ GeV}$	2×6 (J/ψ , topo)	2×6 (J/ψ)	1.4	55
	Two μ , $p_T > 6, 6 \text{ GeV}$, $4.7 < m(\mu, \mu) < 5.9 \text{ GeV}$	2×6 (B , topo)	2×6 (B)	1.4	6
	Two μ , $p_T > 6, 6 \text{ GeV}$, $7 < m(\mu, \mu) < 12 \text{ GeV}$	2×6 (Υ , topo)	2×6 (Υ)	1.2	12
Main Rate				86	1750
B-physics and Light States Rate					200

The complete trigger menu has to respect all technical limitations at L1 and the HLT, which makes the alignment of the physics goals of the menu with the available resources in terms of bandwidth and CPU in the HLT PC farm a challenging task. Recorded events are gathered in data streams, depending on their primary use case and their specific offline reconstruction needs. Figure 1 shows the average recording rate of the physics data stream of all ATLAS pp runs taken in 2018. Events for physics analyses are recorded at an average rate of ~ 1.2 kHz.² This comprises two streams, one dedicated to B -physics and light states (BLS) data, which averaged 200 Hz, and one for main physics data, which averaged 1 kHz. The BLS data are kept separate so the offline reconstruction can be delayed if available resources for offline processing are scarce due to high LHC uptime.

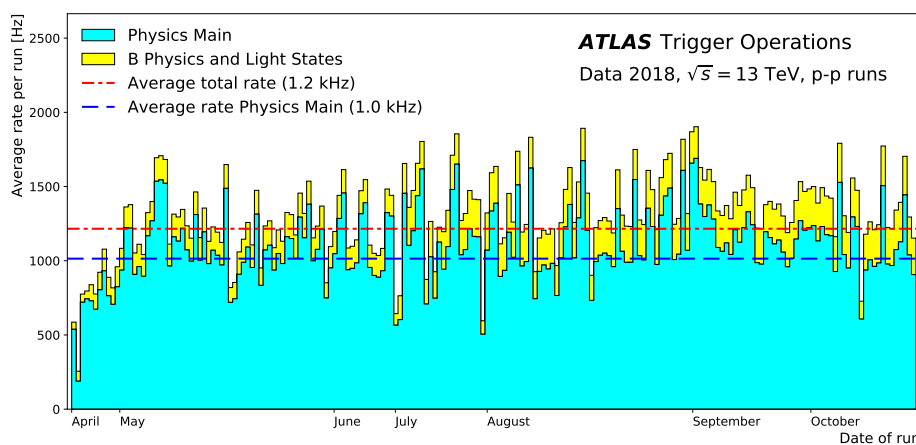


Figure 1: The average recording rate of the main physics data stream and the B -physics and light states data stream for each ATLAS pp physics run taken in 2018. The total average of all runs is indicated as a red dash-dotted line, and the total average of the main physics stream is indicated as a blue dashed line.

The main physics stream contains data from the full detector when a trigger is activated. However, some other streams, such as those used for detector calibration, only record partial information, a strategy called partial event building (PEB). In PEB, only specific parts of the detector or subdetectors are stored, which can reduce the event size by a large factor. An alternative approach to reduce the event size is followed by the trigger-level analysis (TLA) [7, 8], where only the objects reconstructed by the HLT are stored. By reducing the size of the event data, the triggers can operate at much higher rates, without saturation of the output bandwidth. However, other limitations, such as CPU resources and readout rates, prohibit arbitrarily high rates even with minimal partial event information. The smallest event size recorded in 2018 was for AFP calibration data, at 1.3 kB per event, compared to an average full event size of 1 MB per event. Figure 2 shows the HLT rates and output bandwidth as a function of time in a given run. The apparent mismatch between rate and output bandwidth in some streams is due to the use of PEB techniques. The increase of the TLA HLT output is part of the end-of-fill strategy [4] of the

² Due to decreasing luminosity during the run, the average rate is about two thirds of the peak rate.

ATLAS trigger. At the end of the LHC fill, L1 and CPU resources are available to reconstruct and record additional events using lower-threshold TLA triggers. The TLA stream only selects events containing jets and saves the HLT jets and limited event information, leading to a much smaller event size than in regular data-taking. This increases the total HLT output rate, but does not significantly increase the total output bandwidth due to the small size of TLA events

There were several additions on top of the trigger menu used in 2017 [4], and a handful of changes and improvements to triggers used in 2017. The online p_T requirements, also referred to as thresholds, remained the same for nearly all triggers. Several improvements, described in Section 4, were introduced for muon, tau, missing energy, and b -jet triggers. Several combined triggers were added to the menu in order to efficiently select events for analyses with complex event topologies. These triggers typically do not significantly increase the overall trigger rate. Several triggers selecting b -hadron decays to electrons were also implemented.

3.1 Trigger menus for special pp runs

Additional trigger menus were designed to be used during special runs. These menus are adapted to the particular data-taking conditions and physics purpose of each run.

Start-up menu: During the first weeks of 2018 data-taking (17 April–6 May), the LHC initially operated with many fewer bunches than nominal and thus at much lower instantaneous luminosities. The luminosity progressively increased up to the nominal run conditions. This period is referred to as start-up, and a dedicated menu was deployed, with the main focus being the commissioning and validation of new triggers, algorithms, and improvements introduced during the end-of-year shutdown. Dedicated triggers were also included for detector, calibration and performance studies.

van der Meer scan menu: The absolute luminosity calibration of the ATLAS detector is performed via a van der Meer (vdM) scan [9]. During the vdM scan runs, events were recorded based on activity in the MBTS (Minimum Bias Trigger Scintillators) and LUCID (Luminosity measurement using a Cherenkov Integrating Detector) subsystems. The events were recorded to a dedicated stream with PEB at a rate of approximately 20 kHz.

$\beta^* = 90$ m menu: Six days of pp data-taking with $\beta^* = 90$ m took place from 2 - 7 July. The programme was divided in two parts: the first part was for elastic physics with 100 ns bunch spacing for cross-section measurements, while the second part was for diffractive physics with 50 ns bunch spacing mainly for glueball and odderon searches. Triggers dedicated to elastic and diffractive collisions were deployed in both scenarios. The elastic triggers are based on protons detected with ALFA [10] at L1 and are directly recorded without further HLT selection. The diffractive triggers select minimum-bias (MB) events, soft dimuon, diphoton, and dijet events at the HLT, on top of L1 ALFA selections.

900 GeV ALFA menu: During 11 - 14 October, several runs were recorded for the elastic measurement programme of ATLAS with a centre-of-mass energy of 900 GeV. For the measurement of the pp elastic scattering cross-sections, forward protons are tagged by the

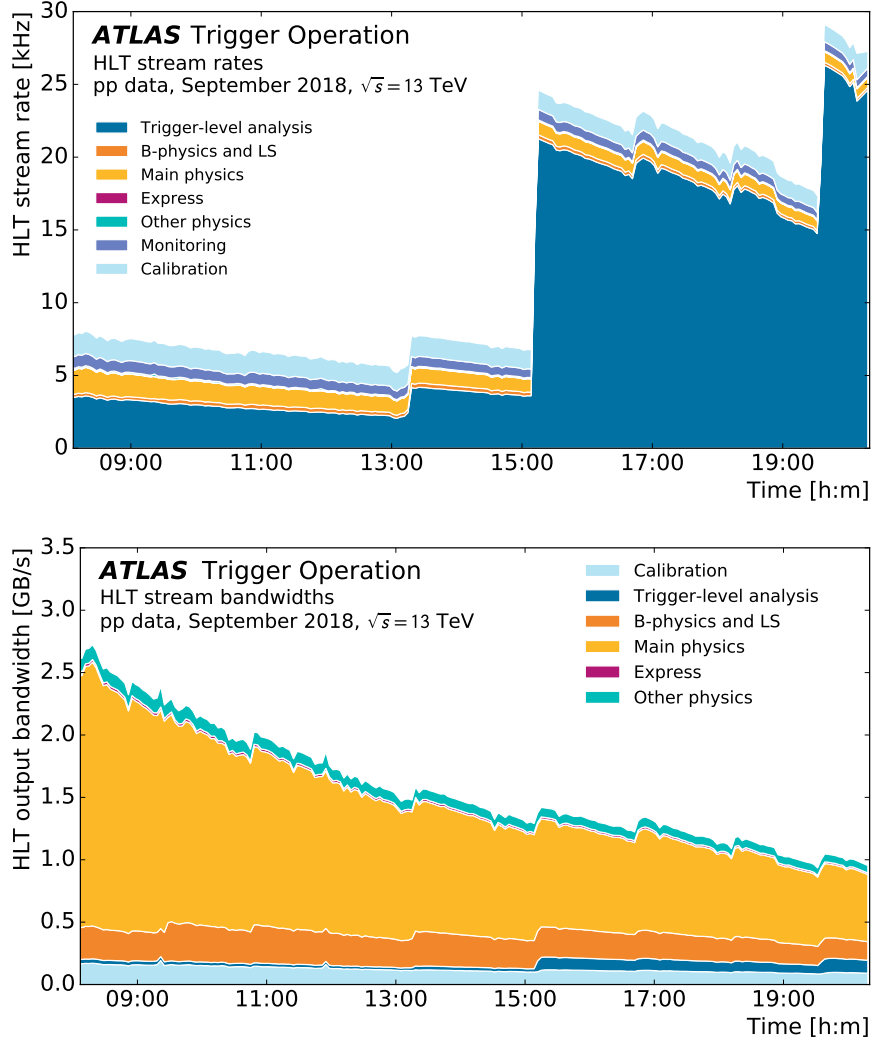


Figure 2: Trigger stream rates (top) and output bandwidth at the HLT (bottom) as a function of time in a fill taken in September 2018 with a peak luminosity of $L = 2.0 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$ and a peak pileup of 56. Presented are the main physics stream rate, containing all triggers for physics analyses; the B -physics and light states (LS) stream, containing triggers specific to B -physics analyses; the express stream, which records events at a low rate for data quality monitoring; other minor streams with physics applications, such as zero-bias and background events; the trigger-level analysis stream; and the detector calibration streams. The monitoring stream is not reflected in the output bandwidth as the monitoring data are not written out to disk.

ALFA detector and recorded without additional HLT selections. Approximately 40 hours of data were recorded.

Low-pileup menu: During 2018, approximately 80 hours of pileup = 2 collisions were recorded on 28 June and 9 - 14 July. There were several physics motivations for this run, including the measurement of $p_T(W)$ which is needed to reduce the error on the measurement of the W mass [11], the heavy-ion long range multi-particle correlation ‘ridge’ analysis [12], E/p measurements and jet calibration, and axion searches. The primary triggers in the low- μ menu were:

W mass: single lepton triggers with a low- p_T threshold, no isolation and loose PID. The HLT threshold was $p_T > 14$ GeV for the single muon trigger and $E_T > 15$ GeV for the single electron trigger. Approximately 40 billion events were recorded by each of these triggers. Additionally, an electron trigger with no PID requirement, HLT threshold of $E_T > 15$ GeV, and HLT missing transverse energy requirement of 30 GeV, intended for W -boson tag-and-probe studies, ran unprescaled.

Ridge studies: a high-multiplicity-track (HMT) trigger with a threshold $n_{\text{tracks}} > 40$, where n_{tracks} stands for a total number of tracks associated to a vertex. The trigger is seeded by a total-energy L1 trigger, for the measurement of azimuthal anisotropies in the angular distributions of particles.

AFP: Triggers based on protons tagged with AFP [13] and jets in ATLAS, with AFP and jet requirements at L1 and jet requirements at HLT.

Axion search: Very low- E_T diphoton triggers ($E_T > 3$ GeV), requiring in addition a veto on total-energy in the detector.

Low-mass dijet resonance search: A single jet trigger with a L1 requirement of $E_T > 20$ GeV and no threshold requirement at HLT was included for a search for low-mass dijet resonances using a TLA [7, 8].

$\langle\mu\rangle = 0.3$ menu: On 26 June and 27 June, two runs with $\langle\mu\rangle = 0.3$ were recorded for both forward physics and angular particle distribution studies. For forward physics, AFP triggers on protons tagged with AFP [13] and jets in ATLAS. Events to study azimuthal anisotropies in the angular distributions of particles were triggered using HMT triggers seeded by total-energy L1 triggers.

4 Improvements and updates to the trigger during 2018

Before and during 2018 data-taking, changes were made to both L1 and HLT triggers. These include improvements to lower output rates without impacting efficiencies and the addition of L1 and HLT triggers to select new topologies.

4.1 Improvements to the L1 trigger

4.1.1 L1 muon trigger

The L1 muon trigger requires spatial and temporal coincidence of hits in resistive plate chambers (RPCs) or thin gap chambers (TGCs) for muons in the barrel ($|\eta| < 1.05$) and endcap ($1.05 < |\eta| < 2.4$), respectively. The muon p_T is estimated by using the degree of deviation from the hit pattern of an infinite momentum assumption. In Run 1, the L1 muon trigger rates in the endcap regions were dominated by low- p_T charged particles originating from low- p_T backgrounds.

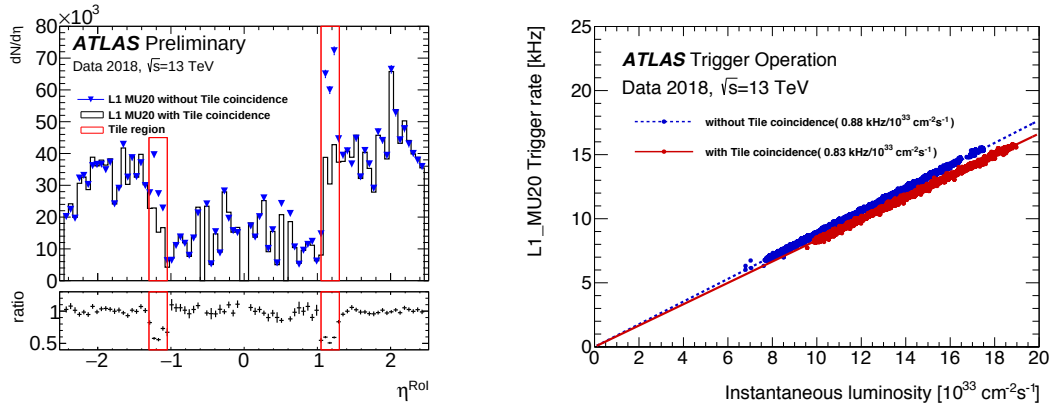


Figure 3: (left) The pseudorapidity distribution of the L1 RoIs (η^{RoI}) that fulfil a 20 GeV requirement (L1 MU20) after the deployment of the new TileCal coincidence in the L1 trigger decisions (white histogram). The L1 MU20 candidates are collected by a pass-through high-level trigger seeded by L1_MU20 in 2018 data with a centre-of-mass energy of 13 TeV and a bunch-crossing interval of 25 ns. The η^{RoI} distribution seen in runs before the deployment of the TileCal coincidence is also shown as a reference (blue triangle) to examine the reduction of the L1 MU20 trigger rate at $1.05 < |\eta^{\text{RoI}}| < 1.3$, which is highlighted by the red rectangles. The reference histogram is normalised so that the entries out of the acceptance of the TileCal coincidence ($1.05 < |\eta^{\text{RoI}}| < 1.3$) are compatible between the two distributions for the comparison. (right) The L1 rate for the single muon trigger with a p_T threshold of 20 GeV as a function of instantaneous luminosity. The red (blue) points correspond to data recorded with (without) the coincidence requirement. The comparison of the fit results shows that the reduction of the L1_MU20 trigger rate for all the muon spectrometer coverage is about 6% owing to the new tile calorimeter coincidence requirement.

To suppress such fake events, an additional coincidence of the TGCs was implemented on 21 May 2018. Muon candidates in the region $1.05 < |\eta| < 1.3$ are outside the coverage of the small TGC wheel, and instead pass through the extended barrel of the tile calorimeter (TileCal). Thus, muons in this region are required to have big-wheel TGC hits coincident with energy deposits of minimum-ionising particles in the outermost layer of the tile calorimeter ($1.05 < |\eta| < 1.3$). The additional coincidence reduces the trigger rate by about 6%, as shown in Figure 3, with signal efficiency losses of 2.5% for muons within $1.05 < |\eta| < 1.3$.

Additionally, the trigger coincidence window (CW) for the endcap region, $1.05 < |\eta| < 2.4$, was optimised for both the 4 GeV (MU4) and 6 GeV (MU6) L1 muon trigger thresholds. This

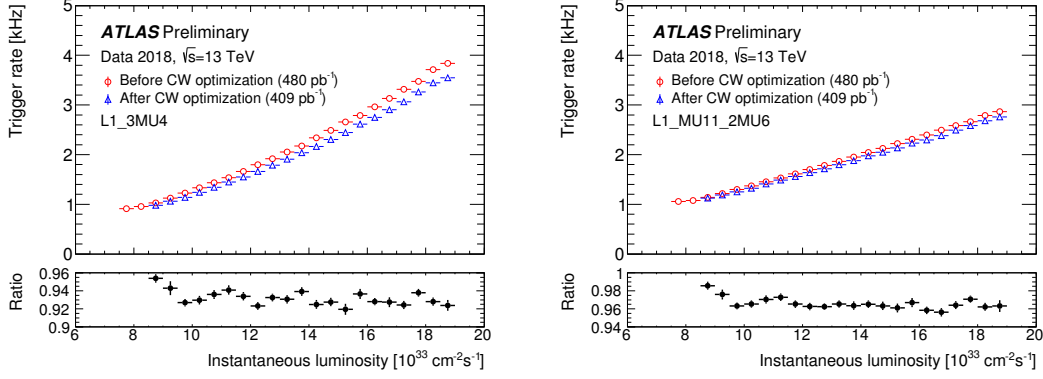


Figure 4: (left) L1 trigger rate for the three-muon trigger with a threshold of 4 GeV, as a function of instantaneous luminosity. (right) L1 trigger rate for the dimuon trigger that requires two muons to satisfy the dimuon trigger with a threshold of 6 GeV and also requires at least one of them to satisfy single-muon trigger with a threshold of 10 GeV, as a function of instantaneous luminosity.

optimisation was deployed at the online ATLAS trigger system on 30 May 2018. This reduced the rate of L1 triggers that include 4 GeV or 6 GeV thresholds; this rate reduction is demonstrated in Figure 4. The convention “MU11” means a single-muon trigger with a threshold of 10 GeV and additional tight requirement (three-station coincidence) in the barrel region ($|\eta| < 1.05$) as compared to the single-muon trigger with a threshold of 10 GeV, which requires only a two-station coincidence in the barrel region (MU10). The CW optimisation reduced the trigger rate for the three-muon trigger with a threshold of 4 GeV by about 7% and the two-muon trigger with a threshold of 6 GeV plus the requirement that at least one muon satisfies MU11 by about 4%.

4.2 Improvements to high-level triggers

4.2.1 Muon

In 2017, the primary single muon trigger demonstrated a decreasing efficiency at high pileup due to the isolation applied to the HLT muons. In order to determine whether or not a muon is isolated, the scalar p_T sum of inner detector (ID) tracks below a specified $\Delta R(\text{track}, \text{muon})$ is calculated. However, since this isolation cone is susceptible to including ID tracks from pileup vertices, an additional requirement is placed on tracks considered for isolation. The difference in longitudinal impact parameters between the track and the muon, $|\Delta z|$, is restricted below a specified value. In 2018, $|\Delta z|$ was lowered from 6 mm to 2 mm, to mitigate a decrease in efficiency seen at high pileup. Since decreasing $|\Delta z|$ means that fewer tracks are used for isolation, the trigger rate did increase. However, a small rate increase was considered acceptable given the corresponding increase in efficiency. The rates and efficiencies for three $|\Delta z|$ requirements are summarised in Table 2. Efficiencies are computed with respect to isolated offline muons that pass a non-isolated HLT trigger with $p_T > 26$ GeV, in simulated $Z \rightarrow \mu\mu$ events.

Table 2: Rates at $2 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$ and efficiencies for two pileup values for the isolated single muon trigger with p_T threshold of 26 GeV, for three $|\Delta z|$ requirements. The requirement used in 2018 was $|\Delta z| < 2 \text{ mm}$.

Requirement on $ \Delta z $	Rate [Hz]	Efficiency at pileup = 60	Efficiency at $\langle \mu \rangle = 80$
$ \Delta z < 6 \text{ mm}$	223	96%	91%
$ \Delta z < 3 \text{ mm}$	232	98%	95%
$ \Delta z < 2 \text{ mm}$	239	99%	97%

4.2.2 Tau

Several improvements were made to tau triggers in 2018, all related to HLT algorithms.

First, the selection of ID tracks pointing to the L1 tau RoI was made more robust at high pileup. The tau trigger *HLT tau25 medium BDT (1-3 fast tracks)* included in the trigger menu in 2017 required the number of tracks found by the Fast Track Finder (FTF) algorithm to be between 1 and 3 [14]. This led to trigger inefficiencies for 3-prong taus at high pileup, due to the increased probability for the FTF algorithm to reconstruct fake tracks at high ID occupancy, in addition to the 3 tau tracks. The problem was mitigated by considering FTF tracks refitted with precision tracking, which reduces the fake track contribution. This tighter track selection was used for the whole 2018 data-taking period - referred to as *HLT tau25 medium BDT (1-3 precision tracks)* trigger in Figure 5. The following two additional HLT algorithm improvements were deployed in July 2018.

An energy calibration based on a Boosted Regression Tree has superseded the simpler calibration that corrects for the calorimeter response and the energy from pileup. This multivariate energy calibration is largely based on that used in the offline tau reconstruction, which slightly improves the trigger efficiency for low- p_T taus. The track association was further tightened to reduce the impact from pileup tracks by reducing the maximum allowed $|\Delta(z)|$ between the leading and sub-leading tracks from 1 mm to 0.75 mm.

An improved tau PID algorithm based on a Recurrent Neural Network (RNN) has superseded the identification algorithm based on a Boosted Decision Tree (BDT). Thanks to the superior jet rejection provided by the RNN algorithm, RNN triggers could be made more efficient for taus than BDT triggers, for equivalent output rates. Dedicated RNN tunes were used for HLT tau candidates with 0, 1, or 2-3 tracks. The RNN trigger is referred to as *HLT tau25 medium RNN (0-3 precision tracks)* in Figure 5.

Figure 5 compares the efficiency of different tau triggers present in the 2018 trigger menu, while Figure 6 shows their online trigger rates.

4.2.3 Transverse missing energy

In 2017 the lowest unscaled transverse missing energy (E_T^{miss}) trigger made use of the *pufit* algorithm that applies a fit to all topological clusters of energy deposits in the calorimeters and

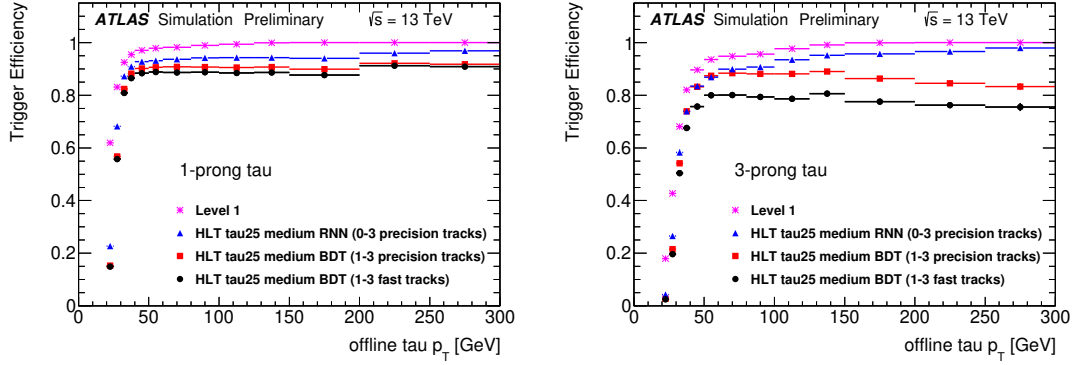


Figure 5: Efficiency of tau triggers as a function of the offline tau transverse momentum p_T , for an E_T threshold of 12 GeV and a medium isolation requirement at L1, and an E_T threshold of 25 GeV at the HLT. The efficiencies are estimated from Monte Carlo simulation using prompt taus produced in $W(\tau\nu)+\text{jets}$ and $t\bar{t}$ events. Efficiencies are computed with respect to 1-prong (left) and 3-prong (right) offline taus passing the medium BDT identification criteria. The three tau trigger versions present in the 2018 trigger menu and the L1 trigger seeding the HLT selections are shown. Only statistical uncertainties are displayed. The three HLT versions have comparable trigger rates.

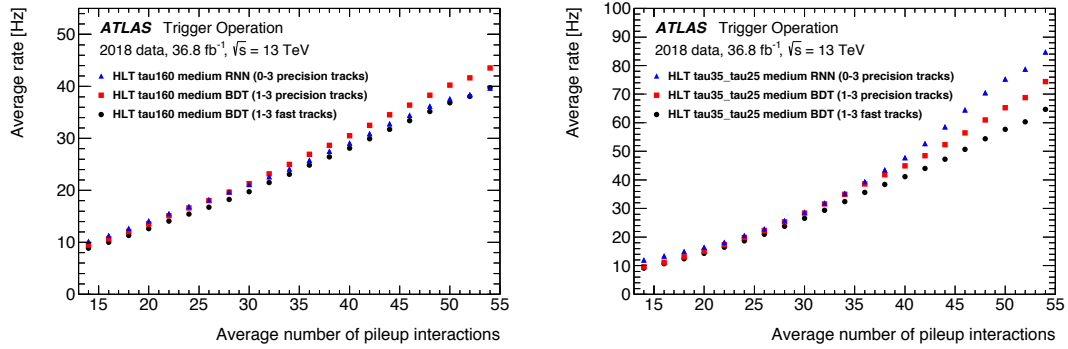


Figure 6: Average rate of tau triggers as a function of the average number of pileup interactions, in pp collisions at $\sqrt{s} = 13$ TeV delivered by the LHC between July and October 2018, corresponding to an integrated luminosity of 36.8 fb^{-1} . Rates are shown for the single-tau trigger with an E_T threshold of 160 GeV at the HLT (left) and the ditau trigger with a L1Topo and a jet requirement at L1, and E_T thresholds of 35 and 25 GeV at the HLT (right). The larger increase in the RNN ditau trigger rate at high pileup comes from HLT tau candidates with no associated track, which were included in the RNN trigger to recover efficiency at low- E_T . The single-tau trigger does not exhibit this behaviour due to its high- E_T threshold.

removes clusters with energy originating from pileup. To reduce the rate but keep the same transverse missing energy threshold at the HLT in 2018 an additional selection on E_T^{miss} calculated by the *cell* algorithm was introduced. At the beginning of 2018 an online *pufit* threshold of 110 GeV in combination with an online *cell* threshold of 70 GeV was chosen to run as lowest unrescaled E_T^{miss} trigger. In May 2018 the *cell* threshold was further reduced to 65 GeV, both triggers ran unrescaled for the rest of the year.

The HLT E_T^{miss} of the *pufit* algorithm is calculated as the negative of the transverse momentum vector sum of all calorimeter topological clusters corrected for pileup. The pileup correction is done by grouping the clusters into coarser “towers” which are then marked as pileup if their E_T falls below a pileup-dependent threshold. A fit to below-threshold towers is performed, taking into account resolutions, making the assumption that the contribution of the pileup to E_T^{miss} is zero. The fitted pileup E_T density is used to correct the above-threshold towers. The second HLT selection in the trigger is on the *cell* algorithm. The E_T^{miss} of this algorithm is calculated as the negative of the transverse momentum vector sum of all calorimeter cells passing a two-sided noise cut.

Detailed studies showed that a large percentage of the online rate is generated by events with comparatively low offline transverse missing energy. These events are mostly not used by physics analyses and rejected by the additional requirement of passing a low *cell* threshold. The combination of the two algorithms exploits the different correlations between online and offline E_T^{miss} of each algorithm. This reduces the E_T^{miss} trigger rate coming from noise and energy mismeasurements while preserving events with genuine E_T^{miss} . The impact of the additional *cell* selection criteria in the plateau region of the trigger efficiency curve is negligible and very small (<1%) in the turn-on region but it reduces the trigger rate by more than 40%.

Figure 7 shows the trigger efficiency as function of the Z boson transverse momentum and the number of pileup interactions for the two lowest threshold unrescaled triggers deployed in 2018. The events in Figure 7 are taken from data with a $Z \rightarrow \mu\mu$ selection and the transverse momentum of the Z boson is used as a proxy for the E_T^{miss} in the event as muons are treated as invisible objects by the triggers concerned.

4.2.4 b -jets

The baseline configuration of b -jet triggers in 2018 used the same flavour-tagging multivariate algorithm tuning that was deployed during the 2017 data-taking period. This was possible due to the general similarity of running conditions between these two years. However, the b -jet trigger menu also introduced several triggers that used a new tuning of the multivariate algorithm intended to increase b -jet flavour tagging efficiency for high- p_T jets (e.g. $p_T > 250$ GeV). This was done using the same tuning procedure adopted in the offline flavour tagging approach and increased the harmonisation between procedures used in the trigger and those used offline.

The baseline training procedure uses a simulation of $pp \rightarrow t\bar{t}$ to develop a BDT (MV2c10) that can provide a measure of the probability that an individual jet arises from the hadronisation of a

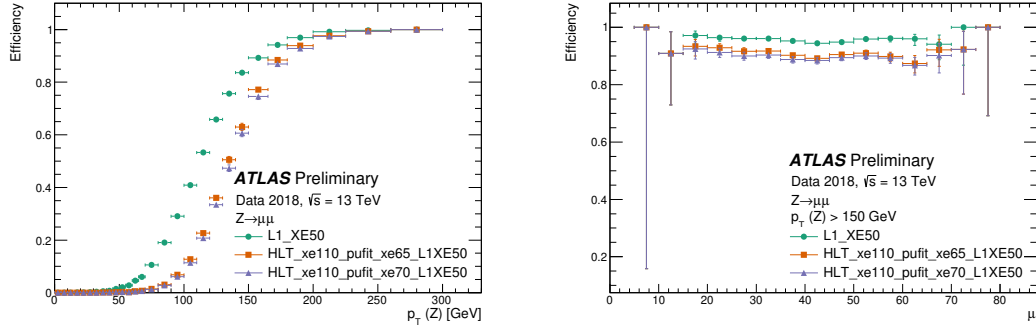


Figure 7: The trigger efficiency of the two lowest unscaled E_T^{miss} triggers as a function of the Z boson transverse momentum (left) and as a function of the number of pileup interaction for events that pass a selection of $p_T(Z) > 150$ GeV (right).

bottom or charm quark, or light quarks and gluons (henceforth referred to as “light” jets). This simulation provides an ample number of jets of various origins to train such an algorithm, but the p_T range is limited due to the kinematics of the underlying physics process. To complement this training sample’s kinematics, an additional simulation sample is introduced into the training procedure, consisting of the process $pp \rightarrow Z' \rightarrow t\bar{t}$, where the heavy vector boson (Z') has a mass of 1 TeV and is produced by the generator with a flat p_T spectrum. These samples are interleaved during training to provide consistent exposure of the BDT to high- and low- p_T jets, and this is referred to as “hybrid tuning”. The b -jet triggers with the new “hybrid tuning” were deployed in August 2018.

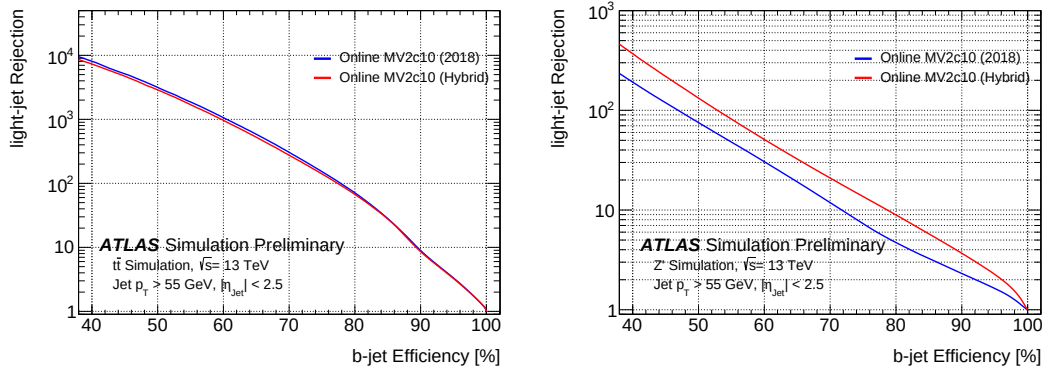


Figure 8: The light-jet rejection vs. the b -jet identification efficiency for the flavour tagging algorithm, measured in simulations of Standard Model $pp \rightarrow t\bar{t}$ (left) and non-Standard Model $pp \rightarrow Z' \rightarrow t\bar{t}$ (right). The MV2c10 algorithm is compared between an online tuning using only $pp \rightarrow t\bar{t}$ simulation in the training procedure (2018), and the online tuning also using the hybrid approach.

The performances of the baseline 2018 online tuning (which uses only $pp \rightarrow t\bar{t}$ simulation in the training), and the hybrid online tuning are compared in Figure 8. It is observed that, as expected, there is little difference in the online 2018 baseline and hybrid approaches in a sample dominated

by low- p_T jets ($t\bar{t}$).

4.2.5 B -physics

Historically, the B -physics trigger programme in ATLAS has focused on decays with muons. In 2018, a new set of triggers was added to select low- p_T dielectron events within an invariant mass region corresponding to decays of b -hadrons. The development focused on $B^0 \rightarrow K^* e^+ e^-$ decays, but all triggers developed are agnostic to additional tracks, and thus are also sensitive to *e.g.* $B^+ \rightarrow K^+ e^+ e^-$ decays. Both L1 items and HLT algorithms were developed.

Mimicking the B -physics L1 muon strategy, L1 topological triggers using EM RoIs were developed. The main trigger requires the invariant mass between two L1 EM RoIs, one with $E_T > 7$ GeV (EM7) and the second with $E_T > 5$ GeV (EM5), to be less than 9 GeV. However, since the granularity of the L1 calorimeter trigger is somewhat coarser than the extent of a single, low- p_T electron, when the two electrons are very close-by they will only create a single L1 EM RoI in the detector. Thus, a L1 trigger for close-by electrons was created, which requires a L1 EM RoI within $\Delta R < 0.3$ of a L1 jet RoI. These two situations are depicted in Figure 9.



Figure 9: Topological sketches of $B \rightarrow K^* e^+ e^-$ decays, where, from the point of view of the L1 EM RoIs, the two electrons are well-separated (left), and the two electrons are overlapping (right). From the centre outwards, the colours represent the inner detector, electromagnetic calorimeter, hadronic calorimeter, and first layer of the muon spectrometer.

On their own, the rates of these two L1 triggers are greater than the global 100 kHz limit, and thus they are combined with a muon requirement to lower the rate: either one MU6 muon or two MU4 muons. Because the production of one b -hadron is likely to result in the production of another, the muon requirement provides a higher purity sample than simply prescaling the original L1 triggers. The rates of the two L1 EM-based triggers with and without combination with the L1 muon selections are shown in Figure 10.

For the L1 trigger with two L1 EM RoIs, two electrons matching the L1 EM RoIs are required in the HLT, with $E_T^1 > 9$ GeV and $E_T^2 > 5$ GeV. For the single L1 EM RoI trigger, two HLT electrons with 5 GeV are required. The HLT electrons were required to pass a very loose electron likelihood (LH) PID [15], without any requirement on the transverse impact parameter. The electrons are required to form a vertex with $\chi^2 < 20$, and the invariant mass of the vertex must be between 0.1 and 6 GeV, which is inclusive of all relevant b -hadron decays. Additionally, HLT muons are

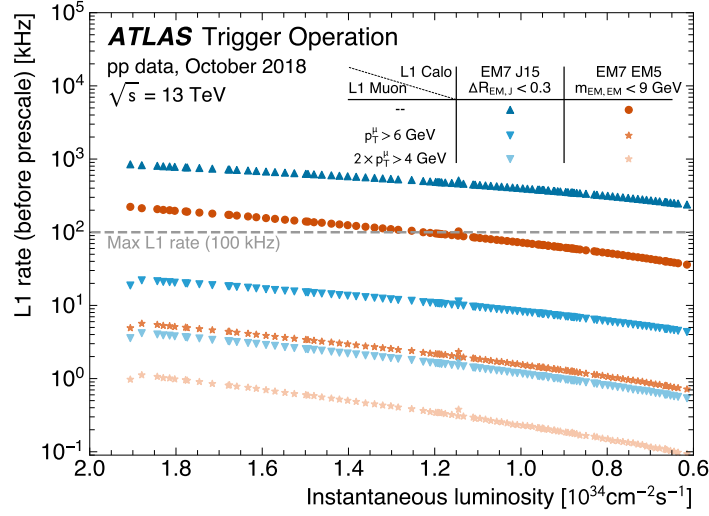


Figure 10: L1 trigger rate, before the application of any prescale, as a function of instantaneous luminosity for the two L1 items targeting $B \rightarrow K^* e^+ e^-$ decays alone and in association with either one 6 GeV muon or two 4 GeV muons. The L1 calorimeter-based thresholds for an EM RoI with $E_T > 5$ GeV, an EM RoI with $E_T > 7$ GeV, and a jet RoI with $E_T > 15$ GeV, are represented by EM5, EM7, and J15, respectively. The four triggers with an L1 muon requirement were deployed online, with end-of-fill prescale strategies.

required in parallel with the L1 muon requirement (either one 6 GeV or 2, 4 GeV HLT muons). These four triggers were deployed on 8 June 2018. An end-of-fill strategy was applied to all triggers except the one with 2, 4 GeV muons and electron thresholds of 9 GeV and 5 GeV, to account for the large L1 rates.

Due to the additional muon requirement, the aforementioned triggers were not fully efficient for signal events, and thus a second set of triggers was developed without the muon requirement. However, as demonstrated in Figure 10, running electron-only items at L1 was not feasible. Thus, instead of requiring dedicated L1 triggers, an HLT trigger was instead run over every event selected by the L1 trigger (L1 accept). HLT electron reconstruction was seeded from EM3 RoIs and two separate triggers were used, one for well-separated electrons and one for close-by electrons. In both cases, two 5 GeV HLT electrons passing a very loose electron LH PID, without any requirement on the transverse impact parameter, are required, with the same vertex requirements as the seeded triggers. It should be noted that, although this trigger ran without an HLT prescale, it only ran if there was an L1 accept from the rest of the trigger menu.

These unseeded triggers were deployed on 14 July 2018, and ran without any HLT prescale below luminosities of $1.85 \times 10^{34} \text{ cm}^{-2} \text{ s}^{-1}$. Above this, small HLT prescales were applied to reduce the excessive pressure on the readout system required to run low- p_T electron reconstruction in every event accepted by the L1 trigger: 5 above $2 \times 10^{34} \text{ cm}^{-2} \text{ s}^{-1}$, 2.5 between $1.95 \times 10^{34} \text{ cm}^{-2} \text{ s}^{-1}$ and $2 \times 10^{34} \text{ cm}^{-2} \text{ s}^{-1}$, and 1.5 between $1.85 \times 10^{34} \text{ cm}^{-2} \text{ s}^{-1}$ and $1.95 \times 10^{34} \text{ cm}^{-2} \text{ s}^{-1}$. These HLT prescales were achieved by online adjustments until the optimal running scenario was

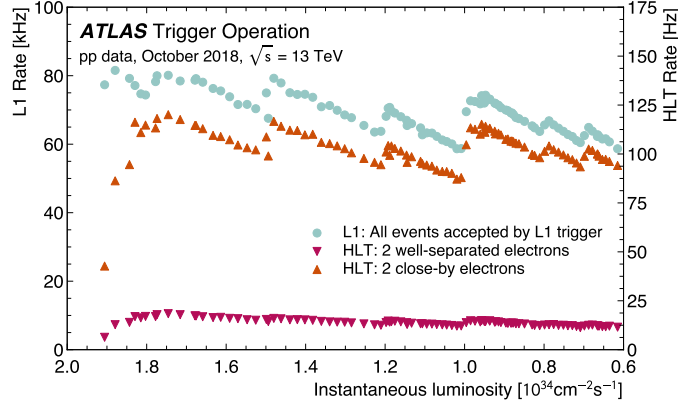


Figure 11: L1 and HLT trigger rates as a function of instantaneous luminosity for the two unseeded triggers targeting $B \rightarrow K^* e^+ e^-$ decays.

achieved. Figure 11 shows the L1 accept rate and HLT rates of each of the two unseeded triggers as a function of instantaneous luminosity, for a run in October 2018.

4.2.6 Dedicated triggers

A selection of triggers targeting specific, signal-driven topologies were added in 2018 to increase the sensitivity of analyses with hard-to-access phase space not recorded by existing triggers.

Photon + multijets: In gauge-mediated Supersymmetry models a possible final state consists of a photon, several jets, and moderate E_T^{miss} . The trigger used for this final state until 2018 was a multijet trigger requiring seven jets with a $p_T > 45$ GeV at HLT. By adding the photon to the final state as a trigger requirement, the number of low p_T jets can be reduced and the jet p_T threshold kept low at the same time. Two triggers were deployed in 2018 for this purpose combining a photon with a p_T threshold of 45 GeV with a requirement of five and six jets with p_T above 45 GeV for primary and support, respectively. The expected gain in signal statistics compared to a pure multijet trigger is expected to be around 30%.

Stopped particle: Several theories of physics beyond the Standard Model (BSM) predict heavy long-lived particles. These particles could be slow-moving and decay in the detector after hours or months. To trigger on these out-of-time decays the ATLAS trigger exploits bunch crossings without pp collisions. A few of these types of triggers have been running online during Run 2. In 2018 an additional trigger was deployed online to improve the number of triggered signal events of the existing triggers. Both the newly introduced and the existing trigger require a central jet with p_T greater than 55 GeV and $E_T^{\text{miss}} > 50$ GeV but the newly introduced one triggers on the five bunch crossings without collisions before and after any bunch crossing with collisions. These bunch crossings are excluded by the existing trigger. Depending on the bunch filling scheme of the LHC the number of additionally

covered bunch crossings varies. With the most conservative estimate based on the number of additional bunch crossings triggered on the expected increase in statistics is in the order of 10%. As these triggers only generate rate for bunch crossings without collisions the rate with the nominal filling scheme is less than 1 Hz despite the comparatively low selection thresholds.

Low-mass diphoton: Several BSM theories predict resonances below the SM Higgs boson mass, for which the diphoton decay mode is the primary mode for discovery. The lowest threshold diphoton trigger running in 2017 required two photons at the HLT with a $E_T > 20$ GeV. However, this trigger is only efficient for $m_{\gamma\gamma} > 60$ GeV, so in order to access these low-mass diphoton states, a dedicated trigger was designed. Lowering the photon E_T thresholds was required to trigger efficiently on the desired phase-space, $10 < m_{\gamma\gamma} < 60$ GeV. However, the standard diphoton trigger with two 20 GeV photons was already the lowest possible threshold, primarily restricted by the large L1 rate of low- E_T L1 EM RoIs. As a compromise, a L1 item with two L1 EM RoIs and a L1 jet RoI was used, with the L1 jet RoI coming from initial-state radiation. With the addition of a jet, the HLT photon E_T threshold was lowered to 6 GeV. The exact L1 jet threshold was dependent on luminosity: 100 GeV above $1.2 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$, 50 GeV with prescale of 20 between $1.0 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$ and $1.2 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$, and 40 GeV below $1.0 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$.

At HLT, two photons passing an E_T threshold of 6 GeV are required. The reconstruction of these photons is seeded from L1 EM RoIs with a p_T threshold of 3 GeV, so if these are not present, HLT photon reconstruction does not proceed. No jet is required in the HLT. Tight PID and isolation are applied to the photons. Additionally, the corresponding triggers with loose photon PID, for performance studies, were added with a constant rate of 0.2 Hz.

Displaced muon: A new single-muon trigger was developed to increase sensitivity to exotic, long-lived neutral particle signatures involving muons produced at a significant distance from the IP. These muons will not produce an ID track, so triggers for these types of signatures use muons with only a muon spectrometer track (MS-only muons). However, the rate of such triggers are incredibly high, especially in the endcaps, and consequently the suitable trigger running in 2015-2017 was restricted to the barrel ($|\eta| < 1.05$, for MS-only muons with $p_T > 60$ GeV). The pseudorapidity restriction results in a much lower global signal acceptance for some models, so in 2018 a new trigger was developed to enable acceptance of endcap MS-only muons. Muons with $1.05 < |\eta| < 2.4$ are required to have at least three good precision layers, an increase from standard muon triggers, which require only one. To ensure the HLT rate was acceptable, the p_T threshold for muons (at all pseudorapidities) is 80 GeV for this trigger, resulting in a rate of 15 Hz at $2 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$. This increased the efficiency for a $Z' \rightarrow \mu\mu$ benchmark signal model with $m_{Z'} = 250$ GeV and $c\tau_{Z'} = 100$ mm from 37% to 54%.

Vector boson fusion: Triggers specifically targeting vector boson fusion (VBF) final states were introduced in July 2018 and collected an integrated luminosity of 41 fb^{-1} . They are all seeded from the new L1Topo trigger MJJ-500-NFF, which requires the invariant mass between two jets to be greater than 500 GeV. Since the invariant mass algorithm requires

a large amount of resources from the L1Topo FPGAs, the invariant mass calculation is only performed on a subset of pairs of L1 jet RoIs. The pairs are formed by selecting one object from each of two lists. The first list is the 6 highest E_T L1 jet RoIs with $|\eta| < 3.2$ and $p_T > 30$ GeV, and the second list contains the 6 highest E_T L1 jet RoIs with $p_T > 20$ GeV, in the full η range ($|\eta| < 4.9$). The topological requirements on the two jets are sufficient to lower the rate enough to unprescale this item at all luminosities. This L1 topological trigger allows the new VBF triggers to have both lower L1 jet E_T thresholds and no L1 requirements on any additional objects in the event.

At HLT, an inclusive trigger broadly targeting all possible VBF final states only places requirements on HLT jets. It requires one $E_T > 70$ GeV jet with $|\eta| < 3.2$ and a second, $E_T > 50$ GeV jet with $|\eta| < 4.9$. Additionally, at least one pair of jets, with each jet satisfying $E_T > 70$ GeV and $|\eta| < 4.9$, must have an invariant mass > 1100 GeV, $\Delta\phi_{jj} < 2.0$, and $\Delta\eta_{jj} > 4.0$. Figure 12 shows the rate of the L1 MJJ-500-NFF item and the efficiency of the inclusive HLT VBF trigger. The efficiency is measured using events that are selected using a single muon trigger with a threshold of 27 GeV. In addition, the events are required to have at least two R=0.4 anti- k_T jets, where one jet has $p_T > 90$ GeV and $|\eta| < 3.2$, while another jet has $p_T > 80$ GeV. The angular requirements $\Delta\phi_{jj} < 2.0$ and $\Delta\eta_{jj} > 4.0$ are also applied offline, as in the trigger requirement.

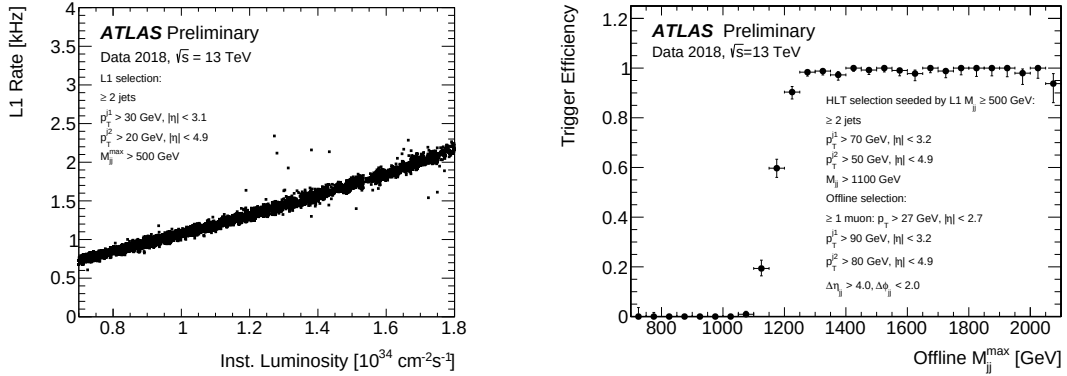


Figure 12: (left) L1 trigger rate for L1_MJJ-500-NFF as a function of the instantaneous luminosity for data recorded in September 2018. Dips are due to dead-time and spikes are caused by detector noise. (right) The trigger efficiency for the inclusive VBF trigger as a function of the offline maximum dijet mass, M_{jj}^{max} .

Additional triggers targeting a large range of final states in association with a high-mass dijet system were created. These triggers are summarised in Table 3.

Table 3: VBF triggers, all seeded by MJJ-500-NFF at L1. All triggers require one, 70 GeV jet with $|\eta| < 3.2$ and a second, 50 GeV jet with $|\eta| < 4.9$, except the dimuon trigger with $2 \times p_T^\mu > 6$ GeV, in which case both jet E_T thresholds are 50 GeV. In all cases the invariant mass is calculated between all pairs of HLT jets in which both jets have $E_T > 50$ GeV and $|\eta| < 4.9$, except the inclusive trigger (as denoted in the table).

Target topology	m_{jj} requirement [GeV]	Additional selection
inclusive	1100	$2 \times p_T^{\text{jet}} > 70$ GeV, $\Delta\phi_{jj} < 2.0$, $\Delta\eta_{jj} > 4.0$
E_T^{miss}	1000	$E_T^{\text{miss}} > 90$ GeV
b -jet	1000	$2 \times p_T^{b\text{-jet}} > 35$ GeV (70% b -jet efficiency)
photon	900	$p_T^\gamma > 35$ GeV (medium PID)
electron	900	$p_T^e > 10$ GeV (medium PID)
muon	900	$p_T^\mu > 10$ GeV
dielectron	900	$2 \times p_T^e > 5$ GeV (medium PID)
dimuon	900	$2 \times p_T^\mu > 6$ GeV
ditau	900	$p_T^\tau > 25, 20$ GeV (medium PID)
electron + E_T^{miss}	1000	$p_T^e > 5$ GeV (very loose PID), $E_T^{\text{miss}} > 50$ GeV
muon + E_T^{miss}	900	$p_T^\mu > 4$ GeV, $E_T^{\text{miss}} > 50$ GeV

Trigger-Level Analysis: The TLA [8] was expanded in 2018 by adding further L1 items seeding the TLA stream. In addition the end-of-fill strategy was optimised by including more luminosity points at which the prescales for the L1 items are reduced. The additional luminosity points exploit the free L1 bandwidth and computing capacity with low luminosities as shown in Figure 13. The TLA stream only selects events containing jets and saves the HLT jets and limited event information, leading to a much smaller event size than in regular data-taking. At the end of the LHC fill, L1 and CPU resources are available to reconstruct and record additional events using lower-threshold TLA triggers and so the prescales of these triggers are reduced or removed at certain luminosity points, as shown by the increases in rate at $L = 1.2, 1.0$, and $0.7 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$. This increases the total HLT output rate, but does not significantly affect the total output bandwidth due to the small size of TLA events.

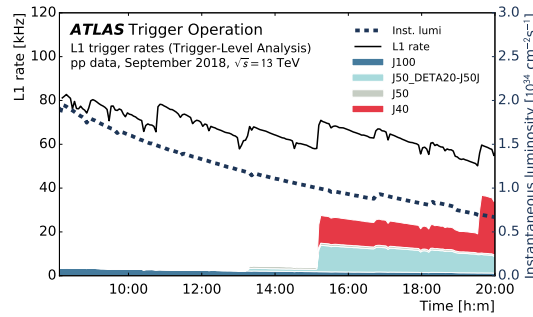


Figure 13: Total L1 trigger output rate as a function of time, with the LHC instantaneous luminosity overlaid, in a fill taken in September 2018 with a peak instantaneous luminosity of $L = 2.0 \times 10^{34} \text{ cm}^{-2}\text{s}^{-1}$ and a peak average number of interactions per crossing of $\mu = 56$. Also shown are the rates of the individual trigger items seeding the TLA stream, without accounting for overlaps between the individual items. The letter “J” followed by a number in the trigger name denotes a jet satisfying a given energy threshold in GeV. The suffix “DETA20-J50J” denotes a leading jet with $E_T > 50 \text{ GeV}$, separated from the sub-leading jet by $\eta < 2.0$. The increase of the L1 rate is part of the end-of-fill strategy of the ATLAS trigger.

Asymmetric dimuon: For analyses looking at a 125 GeV Higgs boson decaying to two (pseudo)scalars and further to two b-jets and two muons the signal efficiency profits from muon trigger thresholds as low as possible. For this purpose an asymmetric dimuon trigger was introduced in 2018 that can be used together with the existing lowest unprescaled single muon trigger. The asymmetric dimuon trigger requires a leading isolated muon with $p_T > 20 \text{ GeV}$ and a second unseeded muon with $p_T > 8 \text{ GeV}$. The muon p_T spectrum depends on the (pseudo)scalar mass and for the cases of soft muons in the final state the inclusion of the asymmetric dimuon trigger gains 2-5% signal efficiency.

5 Trigger menu for 2018 heavy-ion collisions

Heavy-ion (HI) collisions differ significantly from pp collisions. The intrinsic geometry of nuclear collisions results in large event-by-event variations of both the track multiplicity and the energy density. The centrality of Pb+Pb events is defined using the total transverse energy measured in the forward calorimeter, ΣE_T^{FCal} [16]. The ΣE_T^{FCal} distribution is divided into percentiles of the total inelastic cross-section for Pb+Pb collisions, with the 0–10% centrality interval classifying the most central collisions (the highest ΣE_T^{FCal} , the smallest impact parameter which is the distance between the centres of the two colliding nuclei) and the 90–100% classifying the most peripheral collisions (the smallest ΣE_T^{FCal} , the largest impact parameter). The increase in collision centrality results in an increasing number of binary nucleon–nucleon collisions leading to an increase of per-event yields of hard processes. A large amount of additional energy and low momentum particles coming from the underlying event (UE) is produced in more central collisions. Furthermore, the initial spatial asymmetry of the overlap region in the collision of two nuclei are translated into momentum anisotropies of the final-state particle production, which are experimentally characterised by Fourier (flow) harmonics v_n of the azimuthal angle distributions of produced particles [17, 18].

In addition to the collisions with inelastic hadronic interactions, heavy nuclei also interact electromagnetically when the nuclei have significantly larger impact parameters with no hadronic interaction. The cross-section of the electromagnetic interactions, which include photon-photon and photon-nucleus scattering, is enhanced by the large electric charge of Pb nuclei. These collisions, commonly referred to as ultra-peripheral collisions (UPCs) have typically very small track multiplicity and UE. The effective cross-section of electromagnetic processes in Pb+Pb collisions is approximately 200 b for single Pb electromagnetic dissociation [19]. The cross-section for hadronic interactions is 7.7 b [20].

The nucleons that do not interact in the collision of two heavy ions are known as spectators. The Zero Degree Calorimeter (ZDC) [5], located symmetrically at $z = \pm 140$ m and covering $|\eta| > 8.3$, is designed to detect spectator neutrons, which follow a straight trajectory after such collisions. HI triggers use signals from both the main detector and spectator neutrons detected in the ZDC. Furthermore, many triggers are modified compared to their pp counterparts to be able to deal with large event-by-event variations in particle multiplicity.

During the 2018 HI run, the LHC provided Pb+Pb collisions with centre-of-mass energies of $\sqrt{s_{\text{NN}}} = 5.02$ TeV. The instantaneous luminosity reached up to $6.2 \times 10^{27} \text{ cm}^{-2} \text{ s}^{-1}$, leading to a collision rate of 48 kHz for hadronic interactions and approximately 1.5 MHz for events with single Pb electromagnetic dissociation. The rate of accepted events by L1 entering the HLT was restricted by CPU consumption of some triggers and by readout capabilities of several sub-detectors.

The HI trigger menu has the following main components:

- Triggers selecting hard processes including high E_T jets, b -jets, muons, electrons, and photons in inelastic Pb+Pb collisions. The recording rate of these hard probe triggers

decreased with the instantaneous luminosity in a fill. These triggers were recorded in a dedicated hard probes stream.

- Minimum-bias (MB) triggers recording full event information, which are split by centrality to streams for peripheral collisions (PCs) and central collisions (CCs). The recording rate of MB triggers increased as a function of time in a fill.
- MB triggers with PEB, which are split by centrality to streams for PCs and CCs, where the calorimetric information (except FCal) is not recorded. The recording rate of MB triggers increased as a function of time in a fill.
- Triggers collecting events with particular global properties. This includes event-shape triggers collecting events with a large initial spatial asymmetry of the collision and ultra central collision (UCC) triggers selecting the most central HI collisions. The recording rate of these triggers was decreased with the instantaneous luminosity in a fill. Event-shape triggers and UCC triggers were recorded into two independent data streams.
- Triggers selecting various signatures in UPCs. The recording rate of the primary UPC triggers decreased with the instantaneous luminosity in a fill. This decrease in the rate was compensated by a corresponding increase in the recording rate of supporting UPC triggers. All UPC events were recorded to a dedicated UPC stream.

The list of primary triggers included in the 2018 HI trigger menu with their measured rate is presented in Table 4. The number of events recorded by each of these triggers is shown in Figure 14. The average output bandwidth was 4.5 Gb/s through a fill.

Table 4: The main ATLAS triggers used in the 2018 Pb+Pb run with luminosities up to $6.2 \times 10^{27} \text{ cm}^{-2}\text{s}^{-1}$. The observed rates per trigger are reported at a luminosity of $5.0 \times 10^{27} \text{ cm}^{-2}\text{s}^{-1}$ and $2.0 \times 10^{27} \text{ cm}^{-2}\text{s}^{-1}$ as raw rates, including rate overlapping between triggers. The rates of triggers that are not active at a given luminosity are marked by “-” sign. The uncertainty of the stated rates is typically of the order of a few percent. Triggers for signatures marked by $\dagger$ were prescaled during the fill and did not sample the full delivered luminosity. Only a selected number of primary triggers is displayed in this table to indicate the breadth of the physics programme covered by the online trigger selections. An online selection on ΣE_T^{FCal} consistent with noise on the A, C, and both sides, is indicated as FgapA, FgapC, and FgapAC respectively. A requirement on the amount of total transverse energy is indicated as TE. A requirement on a minimum amount of energy in the ZDC consistent with the presence of one or more spectator neutrons on the A, C, and both sides is indicated as ZDC A, ZDC C, and ZDC A & C, respectively. Veto of a requirement is indicated as “!”. PH indicates the number of hits in the pixel detector. More detail information of the selection requirement can be found in sections 5.1-5.5

Trigger	Signature	Trigger Selection		L1	HLT	L1	HLT
		L1 [GeV]	HLT [GeV]	Rate [kHz]	Rate [Hz]	Rate [kHz]	Rate [Hz]
				L= $5.0 \times 10^{27} \text{ cm}^{-2}\text{s}^{-1}$	L= $5.0 \times 10^{27} \text{ cm}^{-2}\text{s}^{-1}$	L= $2.0 \times 10^{27} \text{ cm}^{-2}\text{s}^{-1}$	L= $2.0 \times 10^{27} \text{ cm}^{-2}\text{s}^{-1}$
Leptons	Single μ	6	8	0.47	54	0.21	23
	Two μ	2×4	2×3	0.16	38	0.071	15
	Two $\mu^\dagger$	4	2×4	1.5	28	0.93	15
	Single e (lhloose)	12	15	2.5	11	0.97	4.4
	Single e (loose)	16	20	0.93	35	0.36	14
	Two e (loose)	2×16	2×20	0.29	0.2	0.11	0.13
Photons	Single γ	12	20	2.5	61	0.97	25
Jets	Single jet ($R = 0.4$)	30	85	17.1	120	6.6	47
	Single jet ($R = 0.4$) †	12	60	0.24	13	1.3	69
	Single jet ($R = 1.0$)	50	180	14.4	66	5.6	26
	Single jet ($R = 0.4, \eta > 3.2$)	15	55	12.5	25	9.7	19
b -jets	Jet ($R = 0.4$) & μ	4 (μ), 15 (jet)	4 (μ), 60 (jet)	1.8	7	0.7	3
	Jet ($R = 0.4$) & $\mu^\dagger$	4 (μ)	4 (μ), 50 (jet)	-	-	0.93	6
	Jet ($R = 0.3$) & $\mu^\dagger$	4 (μ)	4 (μ), 40 (jet)	-	-	0.93	6
	Jet ($R = 0.2$) & $\mu^\dagger$	4 (μ)	4 (μ), 30 (jet)	-	-	0.93	6
MB	Very peripheral coll.	TE < 50, ZDC A & C	1 (#tracks)	1.4	95	1.2	261
	Peripheral coll. †	50 < TE < 600	none	0.034	34	0.12	116
	Central coll. †	TE > 600	none	0.029	29	1.2	120
MB PEB	Very peripheral coll.	TE < 50, ZDC A & C	1 (#tracks)	5.4	786	9.6	2048
	Peripheral coll. †	50 < TE < 600	none	0.3	298	0.9	920
	Central coll. †	TE > 600	none	0.8	822	0.23	2251
Global	Ultra central coll.	12000 < TE	4450 (ΣE_T^{FCal})	2.3	81	0.89	32
	Event shape †	600 < TE	top 0.1% v_2 in A/C side	-	-	1.2	0.6
	Event shape †	600 < TE	top 0.1% v_3 in A/C side	-	-	1.3	0.6
UPC	$\gamma + \gamma \rightarrow \gamma + \gamma$	4 < TE < 200 & 1 (γ)	FgapAC & < 15 (PH)	7.7	34	3.1	34
	$\gamma + \gamma \rightarrow \gamma + \gamma$	TE < 50 & 2×1 (γ)	FgapAC & < 15 (PH)	1.8	3	0.66	2.7
	$\gamma + \gamma \rightarrow \mu + \mu$	TE < 50 & 4 (μ)	4 (μ)	0.27	5	0.19	2.1
	$\gamma + \gamma(A) \rightarrow \text{HF}$	TE < 200 & 4 (μ) & (!ZDC A or !ZDC C)	4 (μ)	0.25	1.5	0.19	0.6
	$\gamma + \gamma \rightarrow e + e$	TE < 200 & 7 (e)	12 (e)	0.09	1	0.03	0.5
	$\gamma + \gamma \rightarrow \gamma + \text{jet}$	TE < 200 & 7 (γ)	12 (γ)	0.09	6	0.03	2.4
	$\gamma + A \rightarrow \text{jets}$	5 < TE < 200 & ((ZDC A & !ZDC C) or (ZDC C & !ZDC A))	20 (jet)	0.93	69	0.40	24
	$\gamma + \gamma \rightarrow \text{jets}$	4 < TE < 200 & (!ZDC A & !ZDC C)	15 (jet, $R = 0.4$)	0.65	22	0.74	11
	$\gamma + \gamma \rightarrow \text{jets}$	4 < TE < 200 & (!ZDC A & !ZDC C)	15 (jet, $R = 1.0$)	0.65	36	0.74	20
	high multiplicity	4 < TE < 200 & ZDC A & !ZDC C	FgapC & 35 (#tracks)	0.48	2.4	0.20	0.7
	high multiplicity	4 < TE < 200 & ZDC C & !ZDC A	FgapA & 35 (#tracks)	0.47	2.4	0.20	0.9
	$\gamma + A \rightarrow X^\dagger$	TE < 200 & ((ZDC A & !ZDC C) or (ZDC C & !ZDC A))	1 (#tracks)	0.56	15	0.50	12
	$\gamma + A \rightarrow \text{VM}^\dagger$	TE < 20	1-15 (#tracks)	-	-	16.5	60

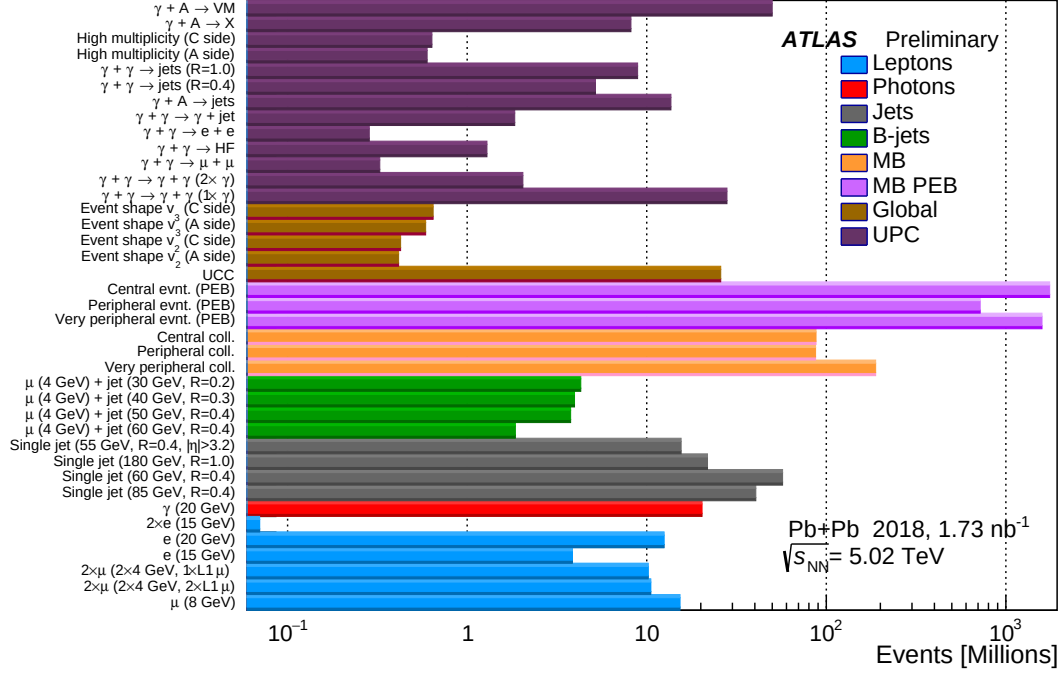


Figure 14: Number of events collected by primary HI triggers during 2018 HI data-taking period. Triggers are grouped into eight categories.

5.1 Minimum-bias and global event properties triggers

Triggers selecting inelastic hadronic events with the smallest bias are called MB triggers. The primary trigger selection of MB HI collisions is based on a logical OR between the following three triggers: 1) a very low activity trigger requiring the ZDC coincidence trigger with a veto on total transverse energy in the calorimeter (TE) to be smaller than 50 GeV at L1 and with a requirement of at least one track reconstructed at HLT; 2) a low activity trigger requiring the TE to be above 50 GeV and smaller than 600 GeV at L1; and 3) a CC trigger requiring at least 600 GeV of total transverse energy at L1. The high-purity central and low activity triggers do not require any additional selection at HLT. The ZDC coincidence trigger is not included in the low activity and central collision triggers, as it is not fully efficient in the most central Pb+Pb collisions where all neutrons from one of the nuclei have interacted leaving none free to be measured by the ZDC.

Two independent MB streams were recorded during the 2018 HI run. The first MB stream collected data from all the sub-detectors. The second MB stream used PEB, and collected data from only the ID, muon spectrometer, ZDC, and FCal. The data from the central calorimeter system were dropped, decreasing the overall event size from ~2200 kB to ~1100 kB for central events and from ~1000 kB to ~200 kB for peripheral collisions and thus allowing for a larger MB dataset to be recorded.

To enhance the number of UCCs recorded, an additional data sample was recorded by dedicated UCC triggers. Three UCC triggers selected events in which the total E_T at L1 is more than 12 TeV and ΣE_T^{FCal} at HLT is in one of three successive intervals with edges 3.61, 4.08, 4.45, and 5.02 TeV. The distribution of ΣE_T^{FCal} with and without weighting by prescale for events sampled by MB and UCC triggers is shown in Figure 15.

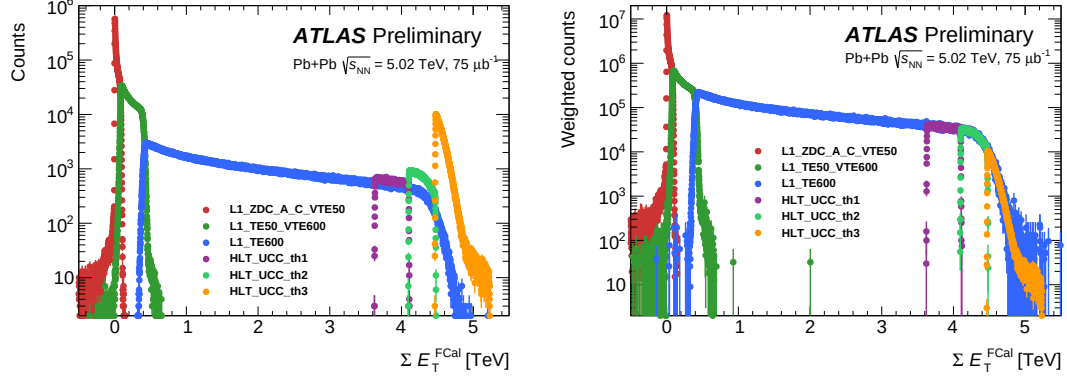


Figure 15: (left) Distribution of ΣE_T^{FCal} for events sampled by three L1 triggers selecting MB events and three UCC triggers. (right) Distribution of ΣE_T^{FCal} for events sampled by MB and UCC triggers weighted by the prescale applied on the corresponding trigger during data-taking. Peripheral collisions are selected by two complementary L1 triggers. The first trigger requires ZDC coincidence (ZDC_A_C) and total energy to be less than 50 GeV (VTE50), and the second peripheral trigger selects events with L1 total energy in the range between 50 and 600 GeV (TE50_VTE600). Central collisions are selected by requiring L1 total energy to be greater than 600 GeV (TE600). Sample of MB collisions is created by combining the sample of peripheral and central events. Sample of the most central collision events is enhanced by UCC triggers with three different thresholds applied at the HLT th1, th2, and th3, corresponding to three successive intervals with edges 3.61, 4.08, 4.45, and 5.02 TeV.

Collective phenomena in heavy-ion collisions are usually inferred from measurements of azimuthal anisotropies of produced particles [21]. To enhance the statistics of central collisions with large azimuthal anisotropies data were collected by several event-shape triggers. These triggers required at least 600 GeV TE at L1 and the top 1%, 2.5%, 5%, and 10% of events with the largest second or third order flow harmonics.

5.2 Jet triggers

Jets are direct probes of the quark-gluon plasma produced in HI collisions, and studying how they are modified in such collisions compared to pp collisions provides insight into their interactions with this QCD medium [22]. The main challenge for triggering on jets in HI collisions is the presence of a large amount of UE. During the 2018 HI run, jet events were identified by L1 jet triggers with thresholds ranging from 12 to 30 GeV without correction for the UE. The HLT jet algorithm used a jet reconstruction procedure similar to that applied in offline analyses [23] and selected events containing jets with transverse energies exceeding a threshold ranging from 50 to

120 GeV. Jets are reconstructed at the HLT across the entire calorimeter using the anti- k_t algorithm with distance parameters $R=0.4$ and 1.0 , from projective towers of size $\Delta\eta \times \Delta\phi = 0.1 \times 0.1$ formed from the summation of calorimeter cell energies. The UE contribution is then estimated and subtracted, event-by-event, from each jet at the calorimeter cell level after the jet finding. The subtraction is performed separately in each η strip of width 0.1 and in each calorimeter sampling layer (i). The background subtracted cell energies (E_{Tj}^{sub}) are calculated according to:

$$E_{Tj}^{sub} = E_{Tj} - A_j \rho_i(\eta_j), \quad (1)$$

where E_{Tj} is the measured cell transverse energy, ρ_i is the average transverse energy density in a given η strip, A_j is the cell area, and j runs over all calorimeter cells. Cells within jet candidates are excluded from the estimate of average UE energy density ρ to avoid biasing the UE estimation. Opposite to the offline algorithm, the UE is not corrected for flow modulation in the trigger.

The jet trigger efficiency with respect to $R = 0.4$ offline jets is presented in Figure 16. The sharpness of the efficiency curves slightly degrades from peripheral to central collisions due to differences in the offline and online jet reconstruction algorithms. The list of primary jet triggers used in the 2018 HI run is presented in Table 4.

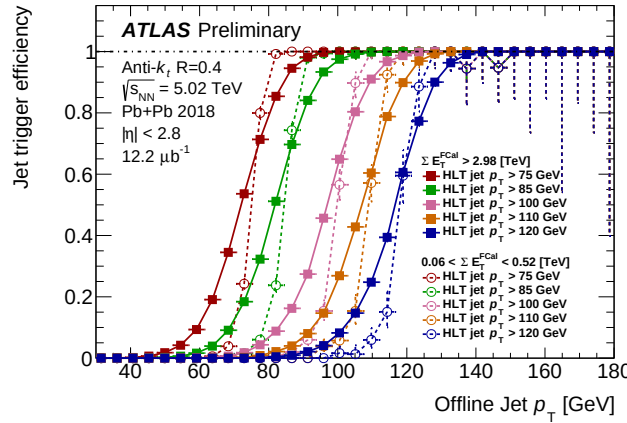


Figure 16: HLT jet trigger efficiency evaluated with respect to offline jets and plotted for different jet trigger thresholds and for two selections of ΣE_T^{FCal} . The efficiency is made with respect to MB triggers. Both the offline and trigger jets are reconstructed using the anti- k_t algorithm with $R = 0.4$. Offline jets are selected in the pseudorapidity range $|\eta| < 2.8$.

5.3 Electron and photon triggers

Prompt photons probe the early stages of the HI collision as, due to their colourless nature, they are transparent to the subsequent evolution of the matter. Complementary information is provided by measuring electrons originating from W and Z boson decays. Furthermore, prompt photons or Z bosons calibrate the recoiling jet in γ -jet and Z -jet measurements that are essential to understand the interactions of colour charges with deconfined matter.

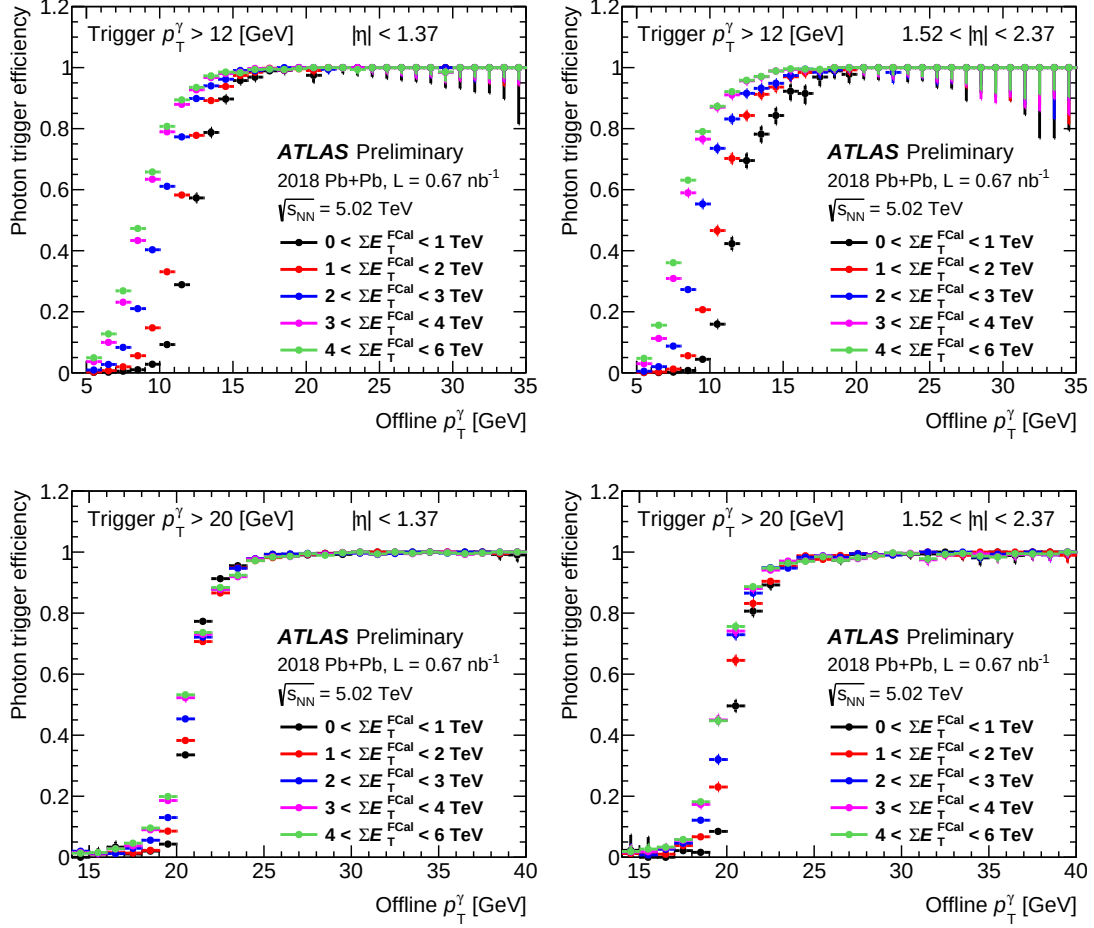


Figure 17: Efficiency for offline photons with tight PID to satisfy the L1 trigger requiring an energy deposit greater than 12 GeV in the EM calorimeter (top) and the HLT trigger requiring the photon to have transverse energy greater than 20 GeV and pass loose PID requirements (bottom) plotted as a function of the p_T of the offline photon. The efficiency is made with respect to MB triggers and is plotted separately for the barrel (left) and endcap (right) regions for five selections of ΣE_T^{FCal} .

Electron and photon signatures are much more robust than jets with respect to the contribution from the UE because they have a physically smaller extent in the detector. Electron and photon candidates are selected with L1 triggers requiring L1 EM RoIs with E_T thresholds of 12 and 16 GeV. To improve the electron and photon trigger performance, a UE subtraction procedure [24] similar to that used in the jet trigger is used at the HLT. Two different electron identification procedures are used: cut-based and LH-based [25] loose PID. The list of primary electron and photon triggers used in the 2018 HI run is presented in Table 4.

The L1 and HLT trigger efficiencies with respect to offline photon candidates with tight PID are shown in Figure 17 separately for the barrel and endcap regions and five different ΣE_T^{FCal} selections. A significant centrality dependence to the photon trigger efficiency is observed at L1 as no correction for UE was applied. The efficiency exhibits only very small centrality dependence at the HLT where UE is corrected for.

Electron triggers use the same L1 seeds as the photon triggers and thus exhibit qualitatively similar behaviour. The HLT efficiency to trigger on offline electrons is shown in Figure 18. Efficiencies for electron triggers requiring either a 15 GeV electron passing a loose LH-based PID or a 20 GeV electron passing a loose cut-based PID are presented. The efficiency after reaching its maximum is higher for the cut-based trigger; however, this gain in efficiency is at the cost of a significant decrease in the purity, as can be seen from the trigger rates listed in Table 4.

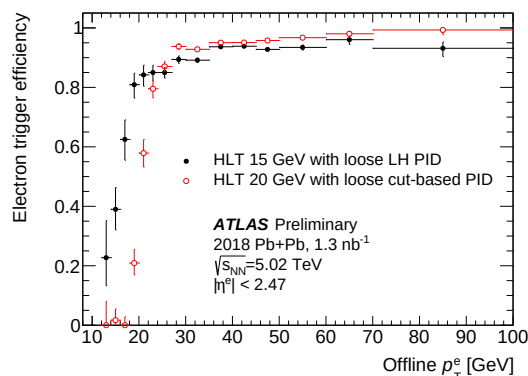


Figure 18: Electron trigger efficiencies as a function of offline electron p_T . The considered triggers require either a 15 GeV electron passing a loose LH-based PID or a 20 GeV electron passing a loose cut-based PID. The efficiencies are measured using the tag-and-probe method in $Z \rightarrow e^+e^-$ candidate events, where probe electrons are required to pass a version of the loose LH-based PID optimised for Pb+Pb collisions.

5.4 Muon triggers

Muon triggers are essential to collect data for measurements of open heavy flavour and quarkonia production, and to collect events with Z bosons decaying to pairs of muons. As the muon triggers are not affected by the presence of UE, no optimisation of the muon trigger algorithm is needed for HI collisions. However, there is a motivation to collect events with muons with the lowest

possible p_T . The lowest threshold of 4 GeV was used to select events containing muon at L1. However, the broad L1 muon trigger turn-on with its $\sim 40\%$ efficiency in the interval 3–4 GeV allows for HLT triggering with p_T thresholds down to 3 GeV. The list of primary muon and dimuon triggers used in the 2018 HI run is presented in Table 4.

5.5 UPC triggers

Various triggers were used in the 2018 HI run to collect data with different signatures occurring in UPCs. A group of triggers was implemented to record events with the light-by-light ($\gamma + \gamma \rightarrow \gamma + \gamma$) scattering process [26]. The effective selection of these events significantly depends on the performance of the L1 calorimeter triggers for low- E_T EM objects and the settings of the system were optimised for low- μ Pb+Pb collisions. The noise settings were tuned to maintain a high efficiency for EM objects down to E_T of 1 GeV and TE triggers with a requirement of 4 GeV for the first time, while maintaining a low rate of noise events.³ Candidate events were selected requiring a logical OR of two conditions at L1: 1) at least one EM RoI with $E_T > 1$ GeV in coincidence with TE between 4 and 200 GeV and 2) at least two EM RoIs with $E_T > 1$ GeV with TE below 50 GeV. The HLT requirement included the ΣE_T^{FCal} to be consistent with noise (FgapAC), and the number of hits in the pixel detector was required to be at most 15. The trigger efficiency for this process was evaluated with $\gamma + \gamma \rightarrow e + e$ events passing an independent UPC trigger, and the resulting L1 efficiency for the pair of clusters is shown in Figure 19. The efficiency is estimated to be greater than 99%.

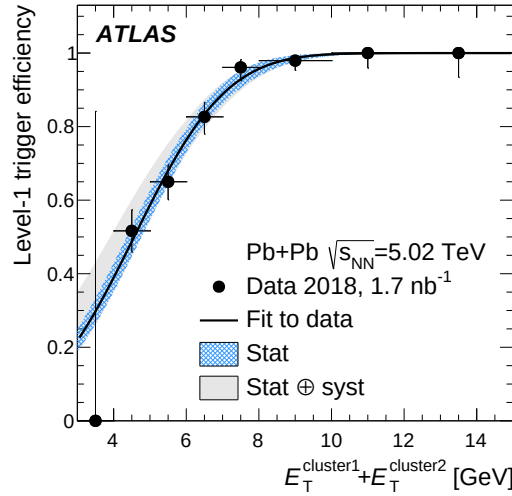


Figure 19: L1 trigger efficiency as a function of the reconstructed energy in $\gamma + \gamma \rightarrow e + e$ events [27].

³ The rate of noise events passing the TE trigger requiring 4 GeV (5 GeV) is estimated to be 9 Hz (1 Hz) for each colliding bunch. The rate of noise events passing L1 trigger requiring an EM object with at least 1 GeV is approximately 0.5 Hz for each colliding bunch.

Dedicated triggers were implemented to record events with the following processes: $\gamma + \gamma \rightarrow \mu + \mu$, $\gamma + \gamma$ (or $+ A$) $\rightarrow$ HF (heavy-flavour production), $\gamma + \gamma \rightarrow e^+ + e^-$, $\gamma + \gamma \rightarrow \gamma + \text{jet}$, $\gamma + A \rightarrow \text{jets}$, and $\gamma + \gamma \rightarrow \text{jets}$. Event candidates for these processes are required to have a maximum TE of 200 GeV at L1 with no signal on both sides of the ZDC for $\gamma + \gamma \rightarrow X$, or with signal on one side and no signal on the other side of ZDC to suppress the contribution of non-diffractive collisions associated with the break-up of both colliding Pb nuclei for $\gamma + A \rightarrow X$ processes. If a muon is required in the final state, the L1 trigger also requires a muon to further reduce the L1 rate. HLT selections require the presence of objects corresponding to the detector signature of each process. More details can be found in Table 4.

A dedicated trigger was introduced to collect UPC data with vector meson (VM) production, which can be used to understand the geometric structure of the nucleus. As different VMs have different binding energies and different effective “sizes”, a trigger was designed to be able to record a variety of VMs. Candidate events were selected requiring less than 20 GeV of TE at L1 and 1–15 tracks at the HLT. Further, a sample of UPC $\gamma + A \rightarrow X$ events was collected with a trigger requiring a maximum TE of 200 GeV, a signal on one side and no signal on the other side in the ZDC at L1, and at least one track at the HLT.

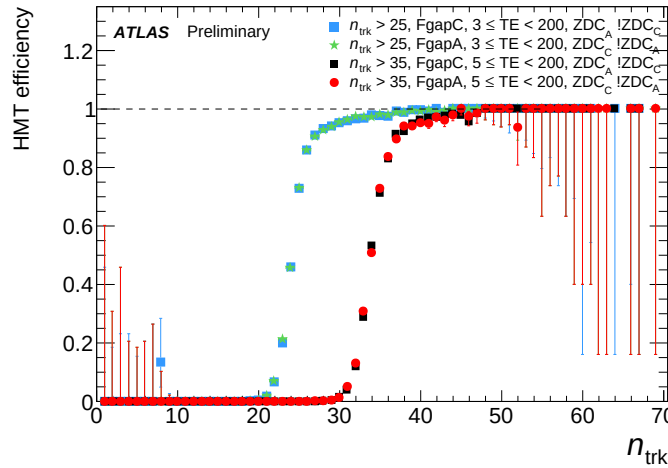


Figure 20: High multiplicity trigger efficiency as a function of offline track multiplicity in photo-nuclear collisions for two different thresholds on a minimum number of charged tracks. The efficiency is shown separately for Pb going to A or C side where a minimum amount of energy in the ZDC consistent with the presence of one or more spectator neutrons is required. The other side is referred to as the photon-going side, where energy deposited in the ZDC is below a maximum-energy cutoff, consistent with no neutrons.

Charged particle multiplicity in photo-nuclear, lower-energy largely asymmetric collisions is typically smaller than in pp collisions. Since the azimuthal anisotropies of produced particles were observed even in pp collisions at the LHC [28], it is a question if such collective behaviour persists in an even smaller collision system. To perform these studies, a dedicated sample of high-multiplicity UPC events was recorded. The trigger selection at L1 includes the requirement of minimal TE ranging from 3 to 5 GeV depending on the target track multiplicity, a requirement

on the maximal TE of 200 GeV, and signal on one side and no signal on the other side of the ZDC. The L1 selection was followed by requiring ΣE_T^{FCal} to be consistent with noise on the side with no ZDC signal (FgapA or FgapC) and at least 15, 25, or 35 tracks, respectively, at the HLT. Trigger efficiency for high multiplicity UPC photo-nuclear collisions as a function of the number of reconstructed charged particles is shown in Figure 20.

6 Summary

The ATLAS trigger system was successfully operated in 2018 during proton–proton and heavy-ion data-taking. The design of the trigger menu, as discussed in this note, enables ATLAS to take data for a broad physics programme as well as for detector monitoring and calibrations. Special run conditions are covered by dedicated trigger menus. The triggers used for heavy-ion data-taking are described in detail including the data-taking strategies for different event characteristics.

The continuous effort to improve the trigger menu together with lower pileup conditions in 2018 allowed for the addition of several new triggers throughout the year targeting previously uncovered phase space. Main improvements compared to 2017 are the enhanced performance of the L1 muon trigger and the better efficiency of several HLT algorithms.

Despite challenging conditions with high luminosities and pileup, the improvements ensured smooth and efficient data-taking during 2018 and a successful conclusion of Run 2.

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